

You are an expert product designer, UX designer, frontend engineer, and startup prototype builder.

Build a complete, polished, interactive frontend prototype for a children's learning startup called:

CURIO

Tagline:

LEARN. PLAY. DO. CREATE.

This website will be used as the LIVE PRODUCT DEMO for a college startup/business competition called "Pitch Perfect".

IMPORTANT:

Build this from scratch.

Do NOT reuse any previous implementation, previous routing, previous mission components, previous mockups, or previous code.

The final product must feel like a real edtech startup prototype that could be shown to investors, parents, schools, and competition judges.

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1. PRODUCT CONCEPT

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CURIO is a personalized learning world for children.

Initial target audience:

Children aged 5–8.

Do NOT build specifically for ages 1–10 in the initial product. The eventual platform may expand to 10, but the MVP is intentionally focused on ages 5–8.

Curio is NOT simply:

- an educational video platform
- a quiz app
- a homework app
- a math game
- an AI chatbot for children

Curio is:

"An interactive learning world where children learn concepts, play challenges, apply them in the real world, and create something of their own."

The core learning loop is:

LEARN → PLAY → DO → CREATE

Example:

LEARN:

Understand why plants need sunlight and water.

PLAY:

Complete an interactive plant experiment.

DO:

Observe a real plant at home.

CREATE:

Design your own plant and explain how it survives.

This loop is the core differentiator of the product.

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2. COMPETITIVE POSITIONING

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The product takes inspiration from principles demonstrated by existing products, but **MUST NOT** copy their branding, UI, content, illustrations, names, or mechanics.

Reference competitors:

Khan Academy
Khan Academy Kids
Duolingo
BYJU'S
Osmo
Prodigy

Their broad strengths:

Khan Academy:

Mastery-based academic learning.

Khan Academy Kids:
Early-childhood educational content.

Duolingo:
Habit formation and gamification.

BYJU'S:
Personalized academic learning.

Osmo:
Physical + digital interaction.

Prodigy:
Game-based academic practice.

Curio's differentiation:

"Learning through exploration, application and creation."

Curio combines:

Academic learning

+

Gamification

+

Real-world activities

+

Creativity

+

Life skills

+

Personalized learning

+

Parent involvement

+

Physical/digital interaction

Do NOT make the homepage claim:
"We are better than Khan Academy."

Instead communicate:

"We don't want to replace what works.
We want to connect the pieces."

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3. BRAND

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Brand name:

CURIO

Tagline:

LEARN. PLAY. DO. CREATE.

Brand personality:

Curious
Playful
Intelligent
Safe
Warm
Exploratory
Premium
Trustworthy

The design must NOT look like a cheap children's website.

Avoid:

- excessive rainbow colors
- childish clip art
- generic stock photos
- excessive gradients
- excessive glassmorphism
- overly cartoonish UI
- clutter
- generic SaaS templates

The visual language should feel like:

Premium modern startup
+
Interactive game world

+

Educational credibility

Use original CSS/SVG/vector-style illustrations or generated visual assets where appropriate.

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4. CORE PRODUCT STRUCTURE

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The child enters a world called:

MY CURIO WORLD

The world contains different locations.

Each location MUST have its own completely different mission.

Locations:

1. 🏪 MARKETPLACE
2. 🧪 SCIENCE LAB
3. 🚀 SPACE STATION
4. 🎨 CREATIVE STUDIO
5. 💡 INVENTOR'S WORKSHOP
6. 🌳 EXPLORER FOREST
7. 📖 STORY VILLAGE

CRITICAL REQUIREMENT:

EVERY LOCATION MUST HAVE A DIFFERENT INTERACTIVE EXPERIENCE.

DO NOT route multiple locations to the same mission.

The following mapping MUST be implemented:

MARKETPLACE

→ Picnic Budget Mission

SCIENCE LAB

→ Plant Growth Experiment

SPACE STATION

→ Planet Selection / Space Logic Mission

CREATIVE STUDIO

→ Creature Creator

INVENTOR'S WORKSHOP

→ Bridge Builder

EXPLORER FOREST

→ Nature Detective

STORY VILLAGE

→ Story Builder

PITCH DEMO

→ Complete Learn → Play → Do → Create experience

There must be NO accidental reuse of the Marketplace mission for another location.

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5. WEBSITE STRUCTURE

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Build these major sections/pages:

1. Landing / Home
2. Learning World
3. Individual Missions
4. Skill Map
5. Parent Dashboard
6. School / Teacher Mode
7. How It Works
8. Safety & Trust
9. Business Model
10. Competitive Differentiation
11. Pitch Demo

Navigation:

CURIO logo

Explore World

How It Works

For Parents

For Schools

Safety

Primary CTA:

TRY A MISSION

Secondary CTA:

PITCH DEMO

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6. LANDING PAGE

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Hero headline:

"Learning isn't a lesson.
It's an adventure."

Supporting text:

"Curio is an interactive learning world where children learn, play, explore real life, and create."

Primary CTA:

Explore the World

Secondary CTA:

Try a Mission

Hero visual:

A beautiful illustrated interactive world containing:

Marketplace
Science Lab
Space Station
Creative Studio
Inventor's Workshop
Explorer Forest
Story Village

The hero should immediately communicate:

"This is a world children explore."

Add subtle animation.

Below hero, show:

LEARN → PLAY → DO → CREATE

With four cards.

LEARN:

Understand a concept.

PLAY:

Practice through an interactive challenge.

DO:

Apply learning in the real world.

CREATE:

Build something using what you learned.

Use animated arrows connecting the four.

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7. LEARNING WORLD
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Create a dedicated "My Curio World" screen.

This should NOT look like a normal dashboard.

It should feel like an explorable map.

Display the seven locations as interactive cards/islands/buildings.

Each location should display:

Name

Icon

Short description

Skills

Progress

Mission status

Example:

 Marketplace

"Where numbers become decisions."

Skills:

Math

Budgeting

Decision Making

Button:

EXPLORE MISSION

When clicked, it MUST open the Marketplace mission ONLY.

Repeat for every location.

IMPORTANT:

Use explicit unique identifiers for every mission.

For example:

marketplaceMission

scienceMission

spaceMission

creativeMission

inventorMission

forestMission

storyMission

pitchDemo

Do not use one generic mission route for all locations.

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8. MARKETPLACE MISSION

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Mission:

"Plan the Perfect Picnic"

Story:

"You have ₹200 to plan a picnic for five friends."

Show a virtual marketplace.

Items:

Sandwich — ₹40

Juice — ₹25

Fruit — ₹30

Chips — ₹20

Cookies — ₹35

Water — ₹15

Allow the user to add/remove items.

Display live:

Total

Remaining Budget

Items Selected

The user must stay within ₹200.

Then introduce a decision:

"You have ₹50 left. What should you do?"

Options:

Buy more food

Save the money

Buy something fun

Donate it

Provide educational feedback.

Mission completion:

MISSION COMPLETE

Skills:

Math
Addition
Subtraction
Budgeting
Decision Making
Financial Literacy

Reward:

+150 XP

Badge:

Money Explorer

This mission must ONLY exist in Marketplace.

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9. SCIENCE LAB MISSION

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Mission:

"Grow a Plant"

Story:

"Maya planted a seed. Can you figure out what the plant needs to grow?"

Create an interactive experiment.

Variables:

Sunlight
Water
Soil
Temperature

Present different plant conditions.

Example:

Plant A:
Enough sunlight

Enough water
Healthy soil

Plant B:
No sunlight
Enough water
Healthy soil

Plant C:
Enough sunlight
Too much water
Healthy soil

Ask:

"Which plant is most likely to grow well?"

Allow selection.

Then create an experiment interface.

The child can change:

Sunlight
Water
Soil

Show an animated simulated plant growing differently based on choices.

Example:

Correct balance:
Healthy growth

Too little water:
Wilting

Too little sunlight:
Slow growth

Too much water:
Root problems

Then explain the cause.

Completion:

MISSION COMPLETE

Skills:

Scientific reasoning

Observation

Cause and effect

Basic biology

Reward:

+150 XP

Badge:

Junior Scientist

This MUST be a completely different UI and interaction from Marketplace.

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10. SPACE STATION MISSION

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Mission:

"Choose the Right Planet"

Story:

"Your research crew needs a planet where humans could potentially establish a scientific research base."

Display 4 fictional planets.

Each planet has:

Temperature

Gravity

Atmosphere

Water availability

Distance

Example:

Planet Nova

Temperature: Moderate

Gravity: Earth-like

Water: Available

Atmosphere: Suitable

Planet Pyra

Temperature: Extremely hot

Gravity: High

Water: Low

Planet Frost

Temperature: Extremely cold

Gravity: Low

Water: Frozen

Planet Terra-X

Temperature: Moderate

Gravity: Moderate

Water: Available

Atmosphere: Thin

Allow the child to inspect planets.

Ask:

"Which planet would you choose?"

After selection, explain why.

Then provide a second mini challenge:

"Your spacecraft has limited energy. Which factor should you prioritize?"

This demonstrates:

Decision making

Data interpretation

Science

Logical reasoning

Completion:

MISSION COMPLETE

+200 XP

Badge:

Space Explorer

This MUST NOT use the Marketplace UI.

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11. CREATIVE STUDIO

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Mission:

"Design Your Own Creature"

The child builds a creature.

Choices:

Environment:

Ocean

Forest

Desert

Arctic

Movement:

Wings

Fins

Legs

Gliding

Food:

Plants

Insects

Fish

Special ability:

Camouflage

Speed
Night vision
Water storage

Generate a visual creature card.

Ask:

"How does your creature survive?"

Give 3 prompts or allow a short answer.

Show:

YOUR CREATION

Name
Environment
Abilities
Food
Survival strategy

Skills:

Creativity
Communication
Design Thinking
Problem Solving

Reward:

+175 XP

Badge:

Creative Inventor

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12. INVENTOR'S WORKSHOP

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Mission:

"Build a Bridge"

The child needs to build a bridge capable of carrying 100 kg.

Available materials:

Wood
Steel
Rope
Stone

Each has:

Cost
Weight
Strength

Give the child a fictional budget.

Allow material selection.

Calculate:

Total Cost
Bridge Strength
Bridge Weight

Then simulate whether the bridge succeeds.

Examples:

Too weak:
Bridge collapses.

Strong enough:
Bridge succeeds.

Over budget:
Mission fails.

Teach:

Engineering thinking
Optimization
Problem solving

Trade-offs

Completion:

+200 XP

Badge:

Master Builder

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13. EXPLORER FOREST

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Mission:

"Become a Nature Detective"

Create an illustrated interactive forest scene.

Clickable objects:

Tree
Bird
Rock
Flower
River
Sun
Insect
Mushroom

Ask the child to identify:

3 living things
2 non-living things
Something that needs sunlight
Something that depends on another living thing

Provide feedback.

Skills:

Observation
Classification

Science
Environmental awareness

Reward:

+150 XP

Badge:

Nature Detective

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14. STORY VILLAGE

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Mission:

"Finish the Story"

Start:

"One morning, Aarav found a mysterious box outside his house..."

Give three choices for what happens next.

Then provide another choice.

Finally ask:

"Write or choose an ending."

Use three required words.

Show the completed story.

Skills:

Reading
Vocabulary
Communication
Creativity
Storytelling

Reward:

+150 XP

Badge:

Story Explorer

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15. PITCH DEMO

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This is the MOST IMPORTANT part of the entire website.

The Pitch Demo MUST NOT open the Marketplace.

It must demonstrate the entire Curio philosophy.

Title:

"Experience Curio"

Subtitle:

"See how one idea becomes a complete learning journey."

Create a guided four-stage experience.

STAGE 1 — LEARN

Topic:

Plants need the right balance of sunlight, water and nutrients.

Show a short interactive explanation.

STAGE 2 — PLAY

Give the user a mini plant experiment.

They choose:

Sunlight
Water
Soil

Show the plant response.

STAGE 3 — DO

Show:

"Your real-world mission"

"Find a plant near you and observe it for three days."

Display:

DAY 1
DAY 2
DAY 3

Allow the user to record/select observations.

This can be simulated.

STAGE 4 — CREATE

Ask:

"Design your own ideal plant."

Allow:

Environment
Water needs
Sunlight needs
Special adaptation

Generate a plant card.

Then show:

YOU DIDN'T JUST LEARN ABOUT PLANTS.

YOU:

LEARNED

PLAYED

DID

CREATED

Completion:

+250 XP

Badge:

Curio Explorer

Skill improvements:

Science +8

Observation +5

Creativity +4

Problem Solving +3

The Pitch Demo should take approximately 2–3 minutes.

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16. GLOBAL USER STATE

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Create a global learner state using React state/context and localStorage if appropriate.

Example:

Child:

Aarav

Age:

7

XP:

1240

Level:
Explorer Level 4

Badges:

Junior Scientist
Money Explorer
Space Explorer
Creative Inventor

Skills:

Math
Science
Reading
Creativity
Problem Solving
Decision Making
Observation
Communication
Financial Literacy

When a mission is completed:

XP increases.

Relevant skill increases.

Badge can unlock.

Mission status changes to completed.

The Learning World should visually reflect completed missions.

Use mock data.

NO backend is required.

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17. SKILL MAP
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Create a beautiful Skill Map.

Do not make it look like a school report card.

Organize skills into:

ACADEMIC

Math
Reading
Science

THINKING

Problem Solving
Critical Thinking
Decision Making
Logical Reasoning

CREATIVE

Creativity
Storytelling
Design
Experimentation

LIFE SKILLS

Financial Literacy
Communication
Collaboration
Environmental Awareness

Show progress visually.

Use:

Progress rings
Bars
Nodes
Badges

Do not claim that Curio can scientifically diagnose personality or intelligence.

Use language such as:

"Skills developed through completed activities."

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18. PARENT DASHBOARD

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Create a realistic Parent Dashboard.

Show:

Child:

Aarav

Age:

7

This Week:

4 missions completed

82 minutes learning

3 real-world activities

5 badges

Skill development:

Science ↑

Math ↑

Creativity ↑

Reading →

Show:

"Strengths this week"

"Science exploration and creative problem-solving are emerging as strengths."

Show:

"Area to support"

"Reading comprehension could benefit from additional practice."

Then:

"Try this at home"

"During your next grocery trip, ask your child to estimate the total cost of five items."

Show recent mission activity.

IMPORTANT:

Parent dashboard should demonstrate VALUE to parents.

Do not make it just a list of scores.

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19. REAL-WORLD MISSIONS

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Create a section called:

"Learning Beyond the Screen"

Show cards:

 Observe a plant for 3 days

 Estimate the cost of five grocery items

 Find five geometric shapes around your home

 Identify three ways to reduce household waste

 Interview a family member

 Build something using household materials

The core message:

"Sometimes the best classroom is the world around you."

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20. PHYSICAL + DIGITAL

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Explain:

"Not everything needs a screen."

Create a visual flow:

DIGITAL
↓
REAL WORLD
↓
DIGITAL

Example:

Learn shapes
↓
Find shapes around your house
↓
Photograph / record them
↓
Create something using those shapes

This communicates the Osmo-inspired principle without copying Osmo.

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21. PERSONALIZATION

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Create a section:

"One child. One learning journey."

Explain that Curio can adapt based on:

Age
Current learning level
Progress
Interests
Mission performance
Activity preferences

Show example:

Aarav, Age 7

Math:

82%

Science:

91%

Reading:

61%

Creativity:

84%

Then:

"Recommended Mission"

Build a Mini Ecosystem

Reason:

"Science is a current strength. This mission gives Aarav a chance to apply it while practicing reading comprehension."

Do not overclaim AI.

If AI is mentioned, describe it as:

"Future personalization engine"

or

"AI-assisted recommendations"

Do not say the AI understands the child's personality or diagnoses learning conditions.

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22. MULTILINGUAL INDIA-FIRST EXPERIENCE
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Include language switcher:

English

हिन्दी

मराठी

Use Indian context naturally:

₹ currency
Indian households
Indian school contexts
Indian examples

Do not make the design stereotypically Indian.

Language selector should visibly work at least for key interface labels.

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23. SCHOOL / TEACHER MODE

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Create a separate Teacher/School section.

Show:

Class Overview

Students
Mission completion
Skill gaps
Recommended activities

Example:

Grade 3A

24 students

Science:
78%

Math:
82%

Reading:
64%

"Reading comprehension is the current class-level opportunity."

Teacher can assign:

Science Mission
Math Mission
Story Mission

This is a prototype.

No backend required.

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24. SAFETY & TRUST

=====

Create a major section:

"Built for Children. Trusted by Parents."

Include:

- ✓ Parent-controlled accounts
- ✓ Child-safe content
- ✓ No targeted advertising to children
- ✓ Minimal data collection
- ✓ Parent visibility
- ✓ Age-appropriate experiences
- ✓ No public child-to-stranger messaging
- ✓ Privacy-first design

IMPORTANT:

Do not claim legal certification.

If mentioning regulations such as COPPA, GDPR-K, or India's DPDP Act, say:

"Designed with child-safety and applicable privacy requirements in mind."

Do not claim compliance unless actually implemented and verified.

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25. GAMIFICATION

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Use:

XP
Badges
Levels
Missions
Unlockable locations
Progress
Collections

But explicitly communicate:

"Rewards encourage learning. They don't replace it."

Do NOT make the interface feel addictive or casino-like.

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26. CREATION GALLERY

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Create a section:

"Don't just answer. Create."

Show example child-created projects:

Math:
"My Mini Shop"

Science:
"My Ecosystem"

Creative:
"My Ocean Creature"

Environment:
"My Eco Home"

Story:
"The Mystery Box"

This demonstrates that Curio measures more than quiz scores.

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27. BUSINESS MODEL

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Create a polished business model section.

FREEMIUM:

Free:

- Selected missions
- Basic learning world
- Basic progress tracking

PREMIUM FAMILY:

Proposed price:
₹299–499/month

Include:

- Full learning world
- Advanced missions
- Parent insights
- Personalized recommendations
- Expanded real-world activities
- Creation projects

SCHOOL:

Annual institutional licensing.

Include:

- Teacher dashboard
- Class analytics
- Assignments
- Curriculum mapping

PARTNERSHIPS:

Potential partnerships with:

- Schools
- Educational institutions
- Publishers
- NGOs

Learning organizations

IMPORTANT:

Label all pricing:

"Illustrative / Proposed"

Do not present it as actual market pricing.

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28. COMPETITIVE DIFFERENTIATION

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Create a professional comparison table.

Competitors:

Khan Academy
Khan Academy Kids
Duolingo
BYJU'S
Osmo
Prodigy
Curio

Dimensions:

Academic Learning
Gamification
Real-World Application
Physical + Digital
Creativity
Life Skills
Parent Insights
Personalization
School Integration

Do not make dishonest claims about competitors.

Use qualitative indicators:

Strong
Moderate

Limited
Core Focus

Position Curio around:

Learning
+
Exploration
+
Application
+
Creation

Headline:

"Curio connects the pieces."

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29. WHY NOW

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Create a section:

"Why Curio?"

Core problem:

Children spend increasing amounts of time with digital content.

Parents want that time to be meaningful.

Schools need engaging supplemental learning.

Traditional educational apps often optimize for:

Content consumption
Practice
Scores

Curio focuses on:

Understanding
Application
Creation

Core statement:

"The gap isn't always knowing something.

It's knowing what to do with it."

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30. ROADMAP

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Show:

PHASE 1

Ages 5–8
Math
Science
Life Skills
Learning World
Real-world missions
Parent dashboard

PHASE 2

English
Creativity
More languages
Teacher dashboard
School pilots

PHASE 3

Ages 8–10
Advanced projects
Collaborative learning
Expanded skill ecosystem

PHASE 4

School partnerships
Regional expansion
More languages
International expansion

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31. FINAL VISION

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Final section:

"Imagine if children stopped asking:

'Do I have to study?'

and started asking:

'What's my next mission?'"

Supporting text:

"Our vision is to make learning something children actively experience — not something they passively consume."

CTA:

ENTER CURIO

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32. RESPONSIVE DESIGN

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The website must work on:

Desktop

Laptop

Tablet

Mobile

Desktop is the priority because this is being demonstrated on a laptop/projector.

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33. TECHNICAL IMPLEMENTATION

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Use:

React

Vite
Modern CSS / Tailwind if appropriate

Use reusable components.

Create clean architecture.

Suggested structure:

```
src/  
  components/  
  pages/  
  missions/  
  data/  
  context/  
  assets/
```

Create individual mission components:

MarketplaceMission
ScienceMission
SpaceMission
CreativeMission
InventorMission
ForestMission
StoryMission
PitchDemo

Do NOT create one generic Mission component that simply changes a title.

Shared components may include:

MissionHeader
XPDisplay
SkillProgress
MissionCompletion
WorldCard
BadgeCard
ProgressBar
ParentInsightCard

Use a central learner state.

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34. ROUTING

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Use explicit routes or an equivalent reliable routing architecture.

Recommended:

/

- /world
- /world/marketplace
- /world/science
- /world/space
- /world/creative
- /world/inventor
- /world/forest
- /world/story
- /skills
- /parents
- /schools
- /safety
- /demo

CRITICAL:

Verify every route independently.

No route may accidentally point to another mission.

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35. ERROR PREVENTION

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Before considering the project complete:

Run:

npm install

Then:

npm run build

Fix EVERY build error.

Then run:

npm run dev

Verify the application loads.

Check browser console for errors.

Fix:

Undefined variables

Missing imports

Broken routes

Missing components

Invalid React state

Broken event handlers

Broken links

CSS overflow

Responsive issues

Do not leave known errors.

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36. FUNCTIONAL TEST CHECKLIST

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You MUST manually test:

HOME

→ loads correctly

EXPLORE WORLD

→ opens world

MARKETPLACE

→ opens Picnic mission

SCIENCE LAB

→ opens Plant experiment

SPACE STATION

→ opens Planet mission

CREATIVE STUDIO

→ opens Creature creator

INVENTOR'S WORKSHOP

→ opens Bridge builder

EXPLORER FOREST

→ opens Nature Detective

STORY VILLAGE

→ opens Story Builder

PITCH DEMO

→ opens the complete Learn → Play → Do → Create experience

Each must be DIFFERENT.

Complete Marketplace mission:

→ XP increases

→ skill updates

→ badge updates

Complete Science mission:

→ Science/Observation updates

Complete Space mission:

→ Reasoning/Decision Making updates

Complete Creative mission:

→ Creativity updates

Parent Dashboard:

→ reflects the user's activity

Skill Map:

→ reflects activity

Back navigation:

→ works

Refresh:

→ works without crashing

Mobile:

→ usable

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37. PITCH DEMO REQUIREMENT

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The website must have a prominent:

PITCH DEMO

button in the navigation.

This is the demo that should be used during the competition.

The pitch flow should be:

1. Problem
2. Curio concept
3. Learn
4. Play
5. Do
6. Create
7. Skill development
8. Parent value
9. Vision

But keep the actual interactive portion around 2–3 minutes.

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38. PERFORMANCE

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Do not make the website unnecessarily heavy.

Avoid huge image files.

Use CSS/SVG where possible.

Lazy-load large assets if necessary.

Animations should remain smooth.

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39. FINAL DESIGN TEST

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Before declaring completion, imagine you are a judge seeing Curio for the first time.

Within 10 seconds, the judge should understand:

WHAT:

A learning world for children.

WHO:

Children ages 5–8, with parents and schools as stakeholders.

WHY:

Children should learn by doing, not only consuming content.

HOW:

Learn → Play → Do → Create.

DIFFERENTIATION:

Real-world application + creation + gamification + personalization.

BUSINESS:

Family subscription + school licensing.

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40. FINAL CRITICAL INSTRUCTION

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DO NOT prioritize the number of pages over product quality.

A smaller number of extremely polished, functional experiences is better than many static pages.

The most important parts of the prototype are:

1. Learning World
2. Unique missions
3. Pitch Demo
4. Parent Dashboard
5. Skill Map
6. Learn → Play → Do → Create philosophy

The website should NOT feel like:

"AI generated a bunch of pages."

It should feel like:

"Someone designed an actual children's learning product."

Build the complete project now.

After implementation, test every route and interaction listed above.

If anything fails, fix it before finishing.

Do not stop at generating files.

Build → Run → Test → Fix → Re-test.